FEEL iQueue
For docker
Installation Manual

Version number： V1.0.0
Release date：2024-09-07

Version change record

project name	version number	change time	operation content	operating personnel
FEEL iQueue Installation Manual	1.0.0	2024-09-07	Init Document	Jaring

Contents

Version change record 2
Contents 3
Forward 4
1. Instructions 4
    1.1 Server hardware configuration requirements 4
    1.2 System deployment architecture diagram 5
2. Preparation 6
    2.1 Port description 8
3. Installation procedure 8
    3.1 Config the NTP time synchronization service 8
    3.2 Install Docker and Docker Compose environments 9
    3.3 Start basic services 11
    3.4 FEEL iQueue environment configuration 13
    3.5 Start FEEL iQueue and its supporting services 14
    3.6 Page login 15
    3.7 Impor Licence 15
4. Backup and recovery of the data 16
5. Start And Restart IQueue Server 18
    5.1 Deployment Package Path 18
    5.2 Start IQueue Server 19
    5.3 Restart IQueue Server 19
    5.4 Restart Linux System 19
6. IQueue config 20
    6.1 Configuration on the system software 20
        6.1.1 Password Config 20
        6.1.2 Counter Config 21
    6.2 Configuration file on linux 22
        6.2.1 Logs 22
        6.2.2 nginx Config 23
        6.2.3 iQueue Config 24
7. Database Support 25
    7.1 LOS Table 25
    7.2 Ticket record table 26
8. Redis Support 26

Forward

Thank you for choosing FEEL iQueue of GRG Banking Equipment Co., Ltd. To facilitate your operations, please read carefully the relevant manuals for this product.

1. Instructions

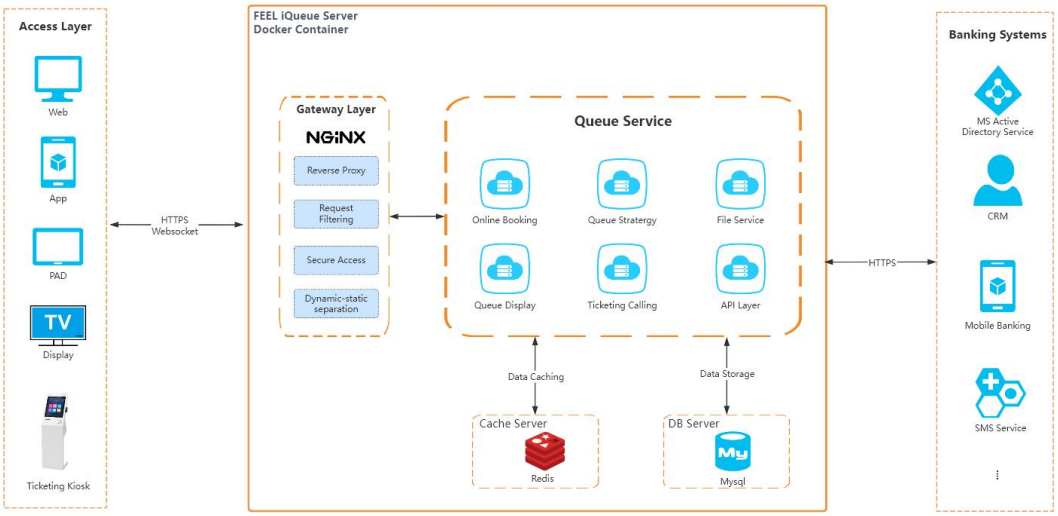
In this document, it descripts the way to install FEEL iQueue product by Docker in Ubuntun OS.

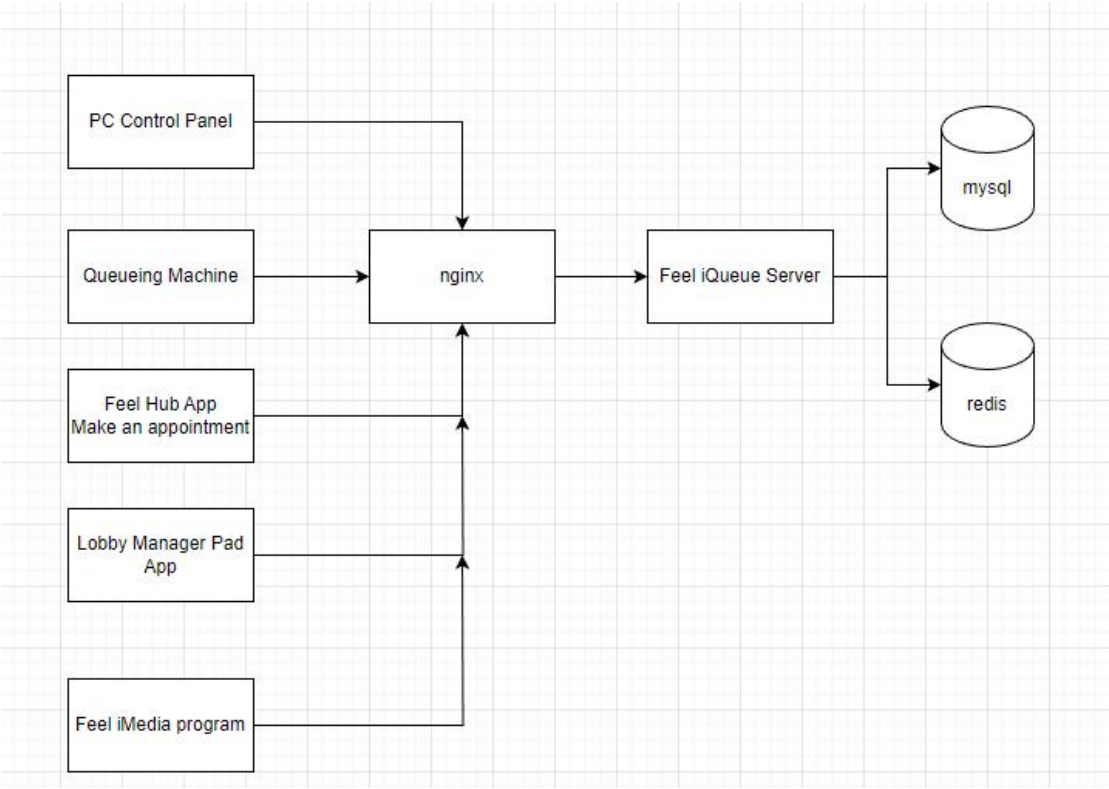
1.1 Server hardware configuration requirements

Hardware requirements		
Processor	CPU core	8 above
	CPU frequency	2.3+ GHz
Memory	Memory type	DDR4 above
	Memory capacity	32G above
Storage	Hard disk size	2TB above (m.2 ssd / hdd)
Network	Gigabit Ethernet	

Software requirements	
Operating system	Ubuntu24 (Priority suggestion)/Centos7
Database	MySQL v8.0.44+
Gateway	Nginx v1.30.2+
Java	Open Jdk 1.8+
Redis	Redis v8.8.1+

1.2 System deployment architecture diagram





1.2.1 System design legend

2. Preparation

Path: /opt

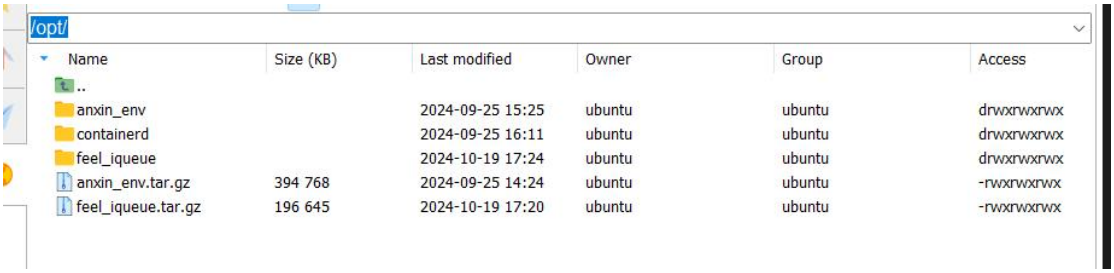
You need to prepare the following files, Upload to the linux server, Decompressing the package

```
cd /opt
users
groups

sudo chown -R ubuntu:ubuntu /opt

tar -zxvf anxin_env.tar.gz
tar -zxvf feel_iqueue.tar.gz
```





ssh tool

名称	修改日期	类型	大小
mysql	2024/10/20 14:55	文件夹	
redis	2024/10/20 14:55	文件夹	
ssh	2024/10/19 17:25	文件夹	
Beyond Compare.zip	2024/9/25 0:34	360压缩 ZIP 文件	9,123 KB

anxin_env.tar.gz：Application environment

/opt/anxin_env/

Name	Size (KB)	Last modified	Owner	Group	Access
..					
dockerImage		2024-09-24 21:56	ubuntu	ubuntu	drwxrwxrwx
install_docker_env		2024-09-25 16:11	ubuntu	ubuntu	drwxrwxrwx
dockerImage.tar.gz	327 055	2024-09-25 00:26	ubuntu	ubuntu	-rwxrwxrwx
install_docker_env.tar.gz	74 289	2024-09-25 00:25	ubuntu	ubuntu	-rwxrwxrwx

install_docker_env.tar.gz：install docker environment

/opt/anxin_env/install_docker_env/

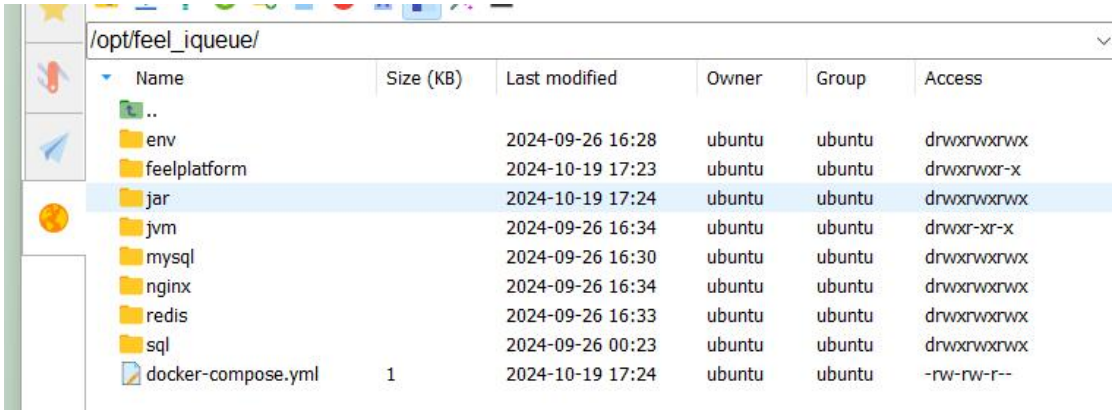
Name	Size (KB)	Last modified	Owner	Group	Access
..					
docker		2022-03-24 09:51	ubuntu	ubuntu	drwxrwxrwx
docker-20.10.14.tgz	62 767	2023-06-07 07:43	ubuntu	ubuntu	-rwxrwxrwx
docker-compose-Linux-x...	11 920	2023-06-07 07:43	ubuntu	ubuntu	-rwxrwxrwx
docker.service	1	2023-06-07 07:43	ubuntu	ubuntu	-rwxrwxrwx
install_env.sh	1	2023-06-07 08:19	ubuntu	ubuntu	-rwxrwxrwx
uninstall_env.sh	1	2023-06-07 07:46	ubuntu	ubuntu	-rwxrwxrwx

dockerImage.tar.gz：docker image

/opt/anxin_env/dockerImage/

Name	Size (KB)	Last modified	Owner	Group	Access
..					
mysql.tar	536 649	2024-09-23 16:58	ubuntu	ubuntu	-rwxrwxrwx
nginx.tar	142 485	2024-09-23 17:00	ubuntu	ubuntu	-rwxrwxrwx
openjdk8.tar	227 025	2024-09-24 21:56	ubuntu	ubuntu	-rwxrwxrwx
redis.tar	113 578	2024-09-23 17:00	ubuntu	ubuntu	-rwxrwxrwx

feel_iqueue.tar.gz：iQueue Application



2.1 Port description

- 6379 redis
- 3306 Mysql
- 8880 map 8880, websocket port of business platform
- 18447 map nginx 443, https communication for web
- 18449 map nginx 443, https communication for pad
- 21581 map 21581, api port of business platform

3. Installation procedure

3.1 Config the NTP time synchronization service

View timedatectl：

```
ubuntu@ubuntu:/opt/feel_iqueue$ timedatectl status
               Local time: Mon 2024-12-02 08:30:41 UTC
           Universal time: Mon 2024-12-02 08:30:41 UTC
                 RTC time: Mon 2024-12-02 08:30:41
                Time zone: Etc/UTC (UTC, +0000)
System clock synchronized: yes
              NTP service: active
          RTC in local TZ: no
```

View time zone:

timedatectl list-timezones |grep "Asia"

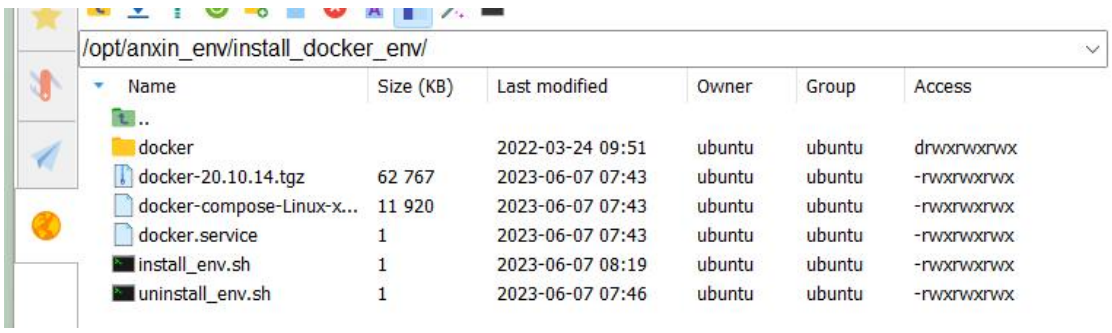
```
ubuntu@ubuntu:/opt/feel_iqueue$ timedatectl list-timezones |grep "Asia"
Asia/Aden
Asia/Almaty
Asia/Amman
Asia/Anadyr
Asia/Aqtau
Asia/Aqtobe
Asia/Ashgabat
Asia/Ashkhabad
Asia/Atyrau
Asia/Baghdad
Asia/Bahrain
Asia/Baku
Asia/Bangkok
Asia/Barnaul
Asia/Beirut
Asia/Bishkek
Asia/Brunei
Asia/Calcutta
```

Select Asia/Hong_Kong or Asia/Shanghai：
timedatectl set-timezone "Asia/Hong_Kong"
timedatectl set-timezone "Asia/Shanghai"

Enable network time synchronization：
Start：
timedatectl set-ntp true

Close：
timedatectl set-ntp false

3.2 Install Docker and Docker Compose environments



- Go to the install_docker_env folder and assign executable permissions to install_env.sh
- Run the following command:

```
sudo chmod 777 install_env.sh
```

./install_env.sh

```
[root@gdyt-lqin1 install_docker_env]# ./install_env.sh
Install docker...
Unzip the docker-20.10.14.tgz package...
docker/
docker/runc
docker/docker-proxy
docker/containerd-shim
docker/docker
docker/ctr
docker/docker-init
docker/dockerd
docker/containerd
docker/containerd-shim-runc-v2
Move the docker directory to /usr/bin...
Move docker.service to the /etc/systemd/system/ directory...
Add file permission...
Reload the configuration file...
Start docker...
Set automatic startup...
Failed to execute operation: Bad message
View docker version
Docker was successfully installed
Docker version 20.10.14, build a224086
Install docker-compose...
Modify docker-compose-linux-x86_64 file name docker-compose...
Move docker-compose into /usr/local/bin...
Add file permission...
View docker-compose version
docker-compose version 1.28.3, build 14736152
Docker-compose was successfully installed
```

```
ubuntu@ubuntu:~$ docker --version
Docker version 20.10.14, build a224086
ubuntu@ubuntu:~$ docker-compose --version
docker-compose version 1.28.3, build 14736152
ubuntu@ubuntu:~$
```

Load Docker Images:

Permissions：
cd /opt
sudo chmod -R 777 ./*

Load mysql：
cd /opt/anxin_env/dockerImage/
sudo docker load -i mysql.tar

Load redis：
cd /opt/anxin_env/dockerImage/
sudo docker load -i redis.tar

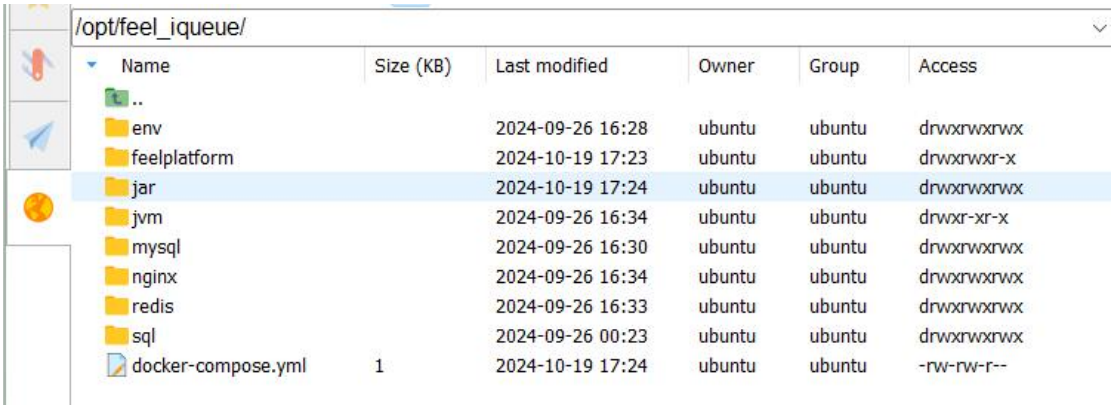
Load nginx:
cd /opt/anxin_env/dockerImage/
sudo docker load -i nginx.tar

Load openjdk:
cd /opt/anxin_env/dockerImage/
sudo docker load -i openjdk8.tar

sudo docker images

```
ubuntu@ubuntu:~$ sudo docker images
REPOSITORY   TAG        IMAGE ID       CREATED         SIZE
openjdk8     openjdk8   4d588cd9ab83   8 months ago    227MB
nginx        latest     605c77e624dd   2 years ago     141MB
redis        latest     7614ae9453d1   2 years ago     113MB
mysql        8.0.21     8e85dd5c3255   4 years ago     544MB
```

3.3 Start basic services



Please enter the following command in the current directory:

Start mysql:
root/Root002152
feelaccess/Feelaccess002152

cd /opt/feel_iqueue/mysql/

View docker status：
sudo docker-compose ps

Docker starts for the first time without starting：

```
ubuntu@ubuntu:/opt/feel_iqueue/mysql$ sudo docker-compose ps
Name   Command   State   Ports
------------------------------
ubuntu@ubuntu:/opt/feel_iqueue/mysql$
```

sudo docker-compose up -d

If Docker has been started, restart：

```
ubuntu@ubuntu:/opt/feel_iqueue/mysql$ sudo docker-compose ps
Name              Command              State                        Ports
-------------------------------------------------------------------------------------------------
mysql   docker-entrypoint.sh --def ...   Up      0.0.0.0:3306->3306/tcp,:::3306->3306/tcp, 33060/tcp
```

sudo docker-compose restart

The Sharding-jdbc table lacks system permissions. Use the mysql tool to execute the following command：
GRANT XA_RECOVER_ADMIN ON *.* TO feelaccess@'%'

Success Result:

```
ubuntu@ubuntu:/opt/feel_iqueue/mysql$ sudo docker-compose ps
Name              Command              State                        Ports
-------------------------------------------------------------------------------------------------
mysql   docker-entrypoint.sh --def ...   Up      0.0.0.0:3306->3306/tcp,:::3306->3306/tcp, 33060/tcp
```

Start redis:
6379: P@ssw0rd
cd /opt/feel_iqueue/redis/

View docker status：
sudo docker-compose ps

Docker starts for the first time without starting：

```
ubuntu@ubuntu:/opt/feel_iqueue/redis$ sudo docker-compose ps
Name   Command   State   Ports
--------------------------------
```

sudo docker-compose up -d

If Docker has been started, restart：

```
ubuntu@ubuntu:/opt/feel_iqueue/redis$ sudo docker-compose ps
   Name                 Command              State                 Ports
-------------------------------------------------------------------------------------------------
queue_redis   docker-entrypoint.sh redis ...   Up      0.0.0.0:6379->6379/tcp,:::6379->6379/tcp
```

sudo docker-compose restart

Success Result:

```
ubuntu@ubuntu:/opt/feel_iqueue/redis$ sudo docker-compose ps
   Name                 Command              State                 Ports
-------------------------------------------------------------------------------------------------
queue_redis   docker-entrypoint.sh redis ...   Up      0.0.0.0:6379->6379/tcp,:::6379->6379/tcp
```

3.4 FEEL iQueue environment configuration

View server IP address

```
ens192: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.1.13.81  netmask 255.255.255.0  broadcast 10.1.13.255
        inet6 fe80::f766:3492:27c7:6d19  prefixlen 64  scopeid 0x20<link>
        inet6 fe80::1be1:1dd:f262:1b84  prefixlen 64  scopeid 0x20<link>
        ether 00:0c:29:62:39:e8  txqueuelen 1000  (Ethernet)
        RX packets 3614304  bytes 4017684366 (3.7 GiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 1700368  bytes 1381300477 (1.2 GiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

Modify the addresses of feel iQueue access mysql , Redis, and file servers



3.3.1 ip address setup Legend

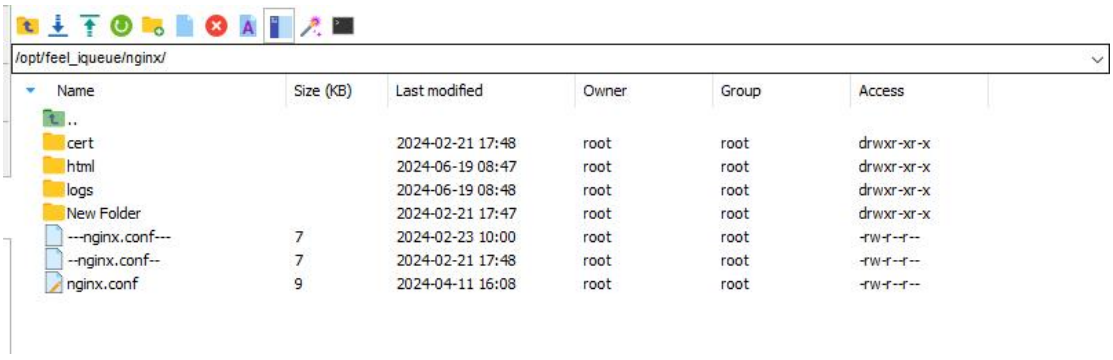
```
setup:
  debug-mode: true # Debug mode switch: common.yml => feel.debug-mode
  apidoc-enable: true # API interface document switch: common.yml => feel.apidoc.enable

  host-redis: 10.1.13.81 # feel_redis # Specify the container name or IP address for Redis: common.yml => redis.host
  host-mysql: 10.1.13.81 # Specify the container name or IP address of MySQL: database.yml => spring.datasource.url

  host-device: 10.252.156.16 # Specify the IP address of the server where the device service layer is located: feelaccess.yml => server.device-service.wsipaddr
  host-fileserver: 10.1.13.81  # Specify the IP address of the server where the file server is located: feelaccess.yml => server.file-service.fsipaddr

  enable-face: 1 # 0-Turn off pixel face service, 1-turn on pixel face service: feelaccess.yml => face.canGetFeature
  host-face: 10.252.156.11 # Specify the IP address of the server where the pixel facial service is located: feelaccess.yml => face.userProfileServerSite
```

3.3.2 mysql redis and fileServer setup Legend



3.3.3 nginx setup legend

Please change the IP address in the configuration file to the actual machine's IP address:

```
        index index.html index.htm;
        try_files $uri $uri/ /index.html;


        location ^~/feel_apis/ {
            proxy_pass https://10.1.13.38:21581/;

            proxy_set_header Host $proxy_host;
            proxy_read_timeout 300;
            proxy_connect_timeout 300;
            proxy_send_timeout 300;
            proxy_set_header X-Real-IP $remote_addr;
            proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;

            proxy_ssl_session_reuse on;
            proxy_ssl_protocols TLSv1.2;
            proxy_ssl_ciphers HIGH:!aNULL:!MD5;
            # 如果后端服务使用自签名证书，可以关闭验证
            proxy_ssl_verify off;
            #确保使用 HTTP/1.1
            proxy_http_version 1.1;
            proxy_set_header Connection "";
            #proxy_set_header X-Forwarded-Proto $scheme
        }

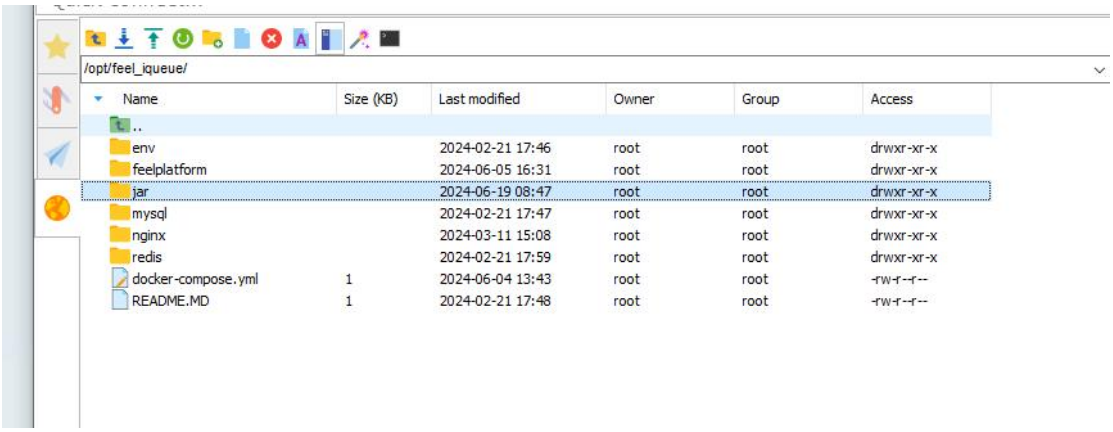
        location ^~/feel_websocket/ {
            # 转发到  业务层(首页)websocket
            proxy_pass https://10.1.13.38:8880/;

            proxy_http_version 1.1;
            proxy_ssl_server_name on;
            proxy_ssl_verify off;
            proxy_set_header Upgrade $http_upgrade;
            proxy_set_header Connection "Upgrade";
            proxy_set_header Remote_addr $remote_addr;
            proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
            proxy_read_timeout 600s;
        }
```

3.3.4 nginx detail setup legend

Note: Please change the IP addresses monitored by ports 443 and 447 to local IP addresses

3.5 Start FEEL iQueue and its supporting services



3.4.1 FEEL iQueue Directory Legend

Start application： cd /opt/feel_iqueue/ View docker status：

sudo docker-compose ps

Docker starts for the first time without starting：

```
ubuntu@ubuntu:/opt/feel_iqueue$ sudo docker-compose ps
Name   Command   State   Ports
-------------------------------
```

sudo docker-compose up -d

If Docker has been started, restart：

```
ubuntu@ubuntu:/opt/feel_iqueue$ sudo docker-compose ps
   Name                 Command              State                                                  Ports
-------------------------------------------------------------------------------------------------------------------------------------------
feel_iqueue   /__cacert_entrypoint.sh ja ...   Up      0.0.0.0:21581->21581/tcp,:::21581->21581/tcp, 0.0.0.0:8880->8880/tcp,:::8880->8880/tcp
queue-web     /docker-entrypoint.sh ngin ...   Up      0.0.0.0:18447->443/tcp,:::18447->443/tcp, 80/tcp
```

sudo docker-compose restart

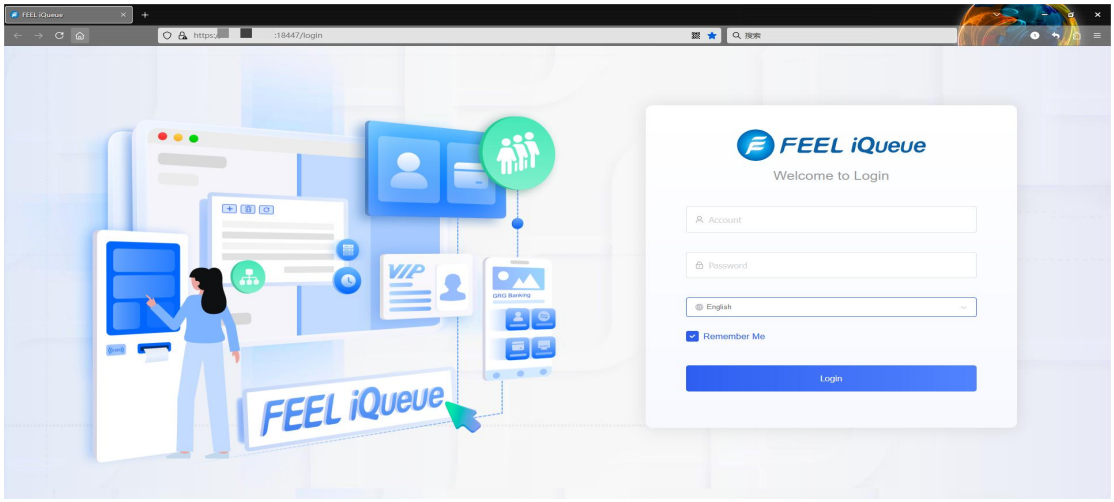
Success Result:

```
ubuntu@ubuntu:/opt/feel_iqueue$ sudo docker-compose ps
   Name                 Command              State                                                  Ports
-------------------------------------------------------------------------------------------------------------------------------------------
feel_iqueue   /__cacert_entrypoint.sh ja ...   Up      0.0.0.0:21581->21581/tcp,:::21581->21581/tcp, 0.0.0.0:8880->8880/tcp,:::8880->8880/tcp
queue-web     /docker-entrypoint.sh ngin ...   Up      0.0.0.0:18447->443/tcp,:::18447->443/tcp, 80/tcp
```

3.4.2 Start Service Legend

3.6 Page login

Use Google browser visit https://[ip address]:18447/login indicates the ip address of the Linux serve



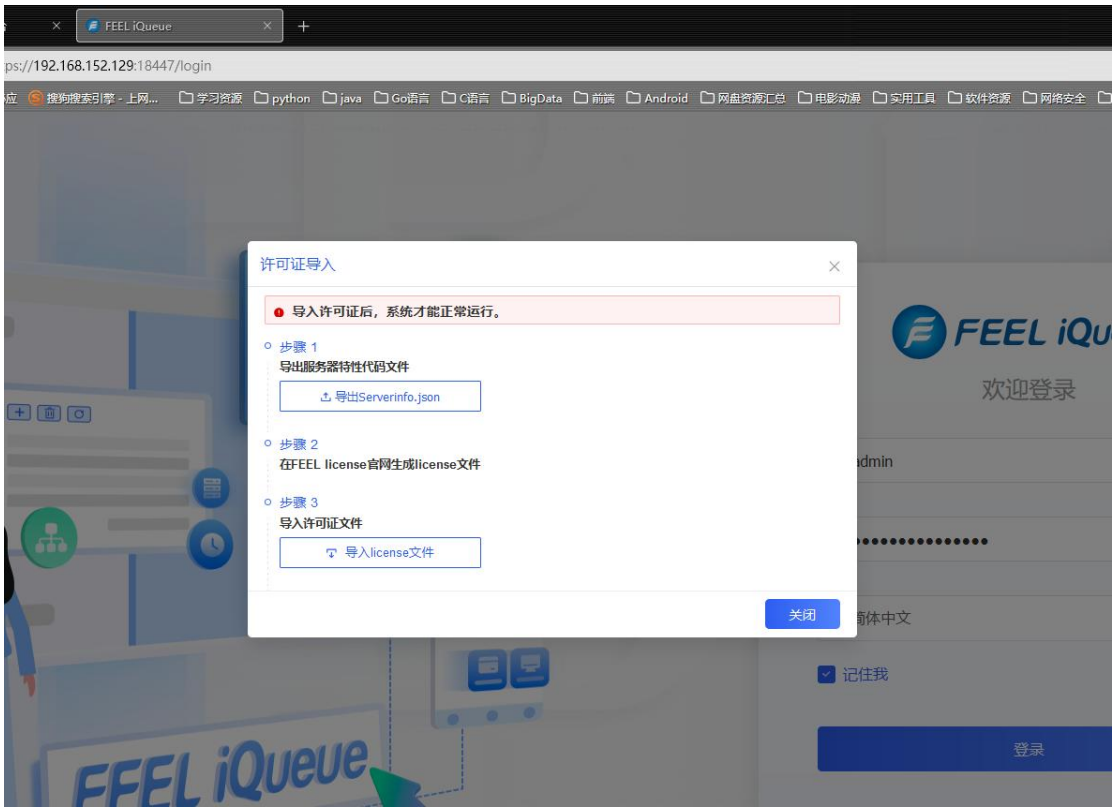
3.5.1 login page Legend

3.7 Impor Licence

Export Serverinfo.json

Apply for a licence from the relevant personnel

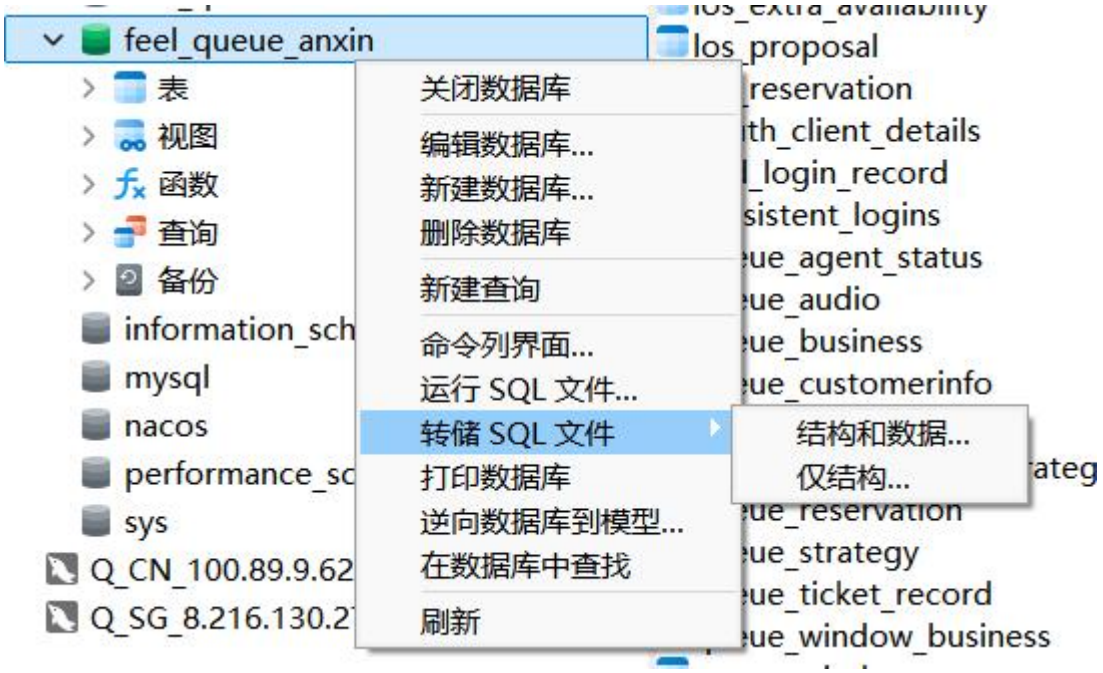
Import licence

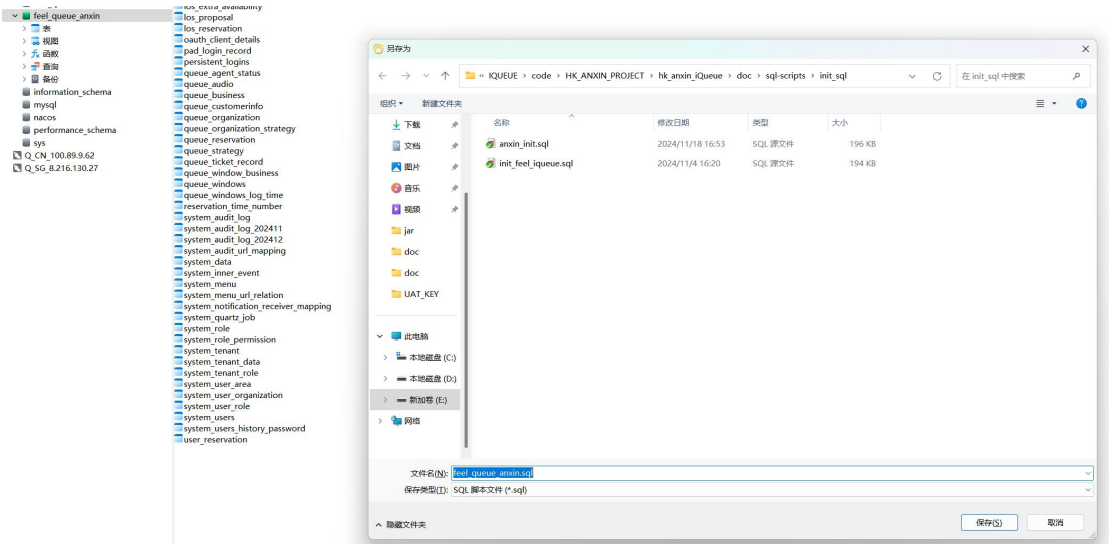


4. Backup and recovery of the data

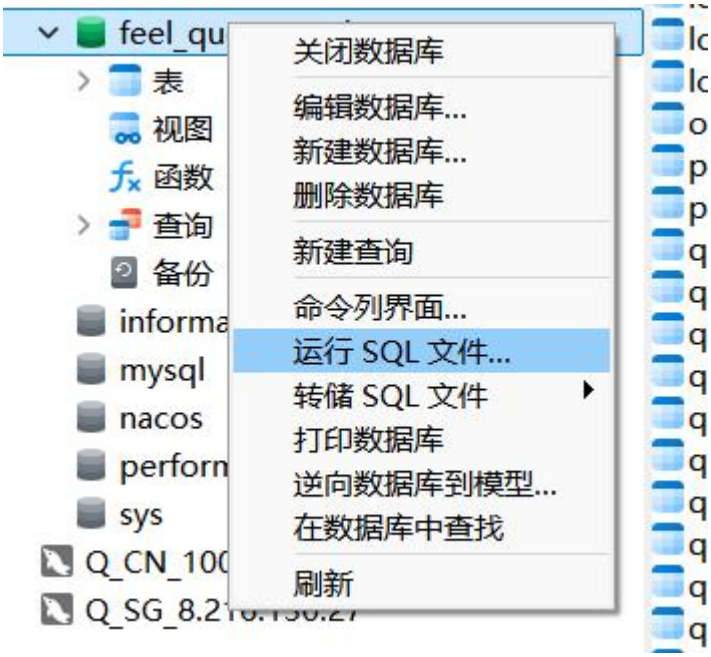
Software: DBeaver 24.2.0 / Nvicat / MySql WorkBench 8.0/ other Mysql tools

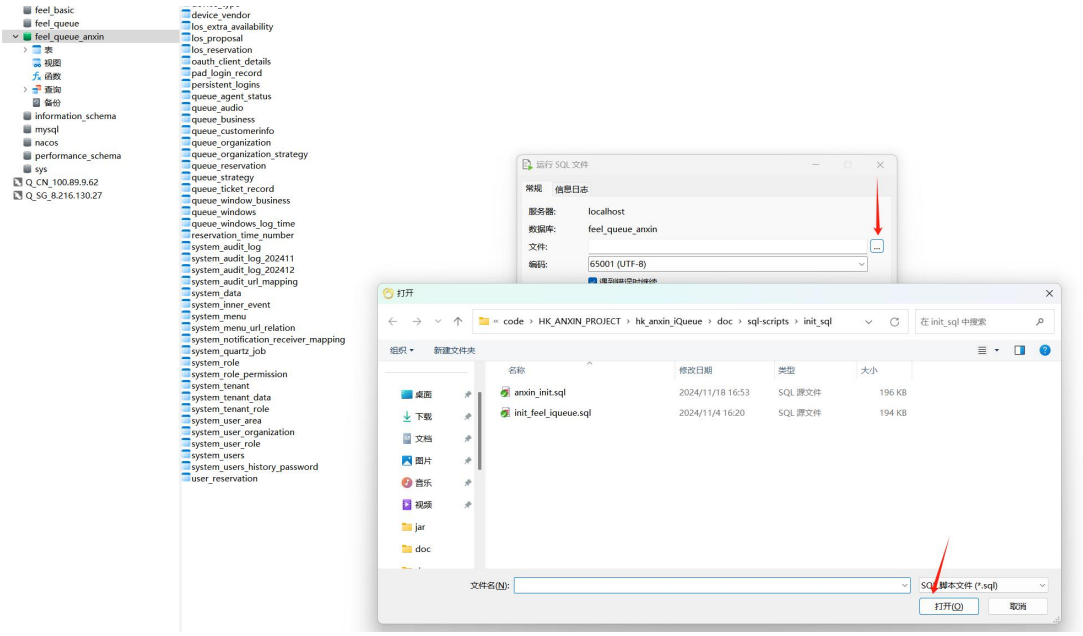
Dump Database:





Restore Database:

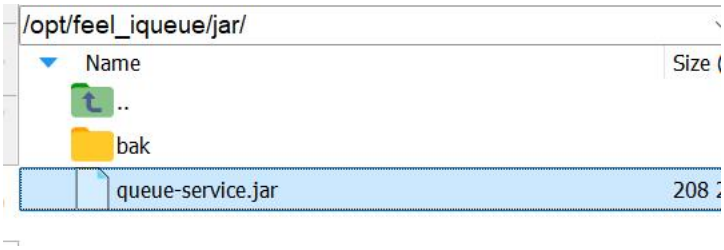




5. Start And Restart IQueue Server

5.1 Deployment Package Path

Back-end package path:：
/opt/feel_iqueue/jar/queue-service.jar



Front-end packet path：
/opt/feel_iqueue/nginx/html/dist



5.2 Start IQueue Server

cd /opt/feel_iqueue/

View docker status：
sudo docker-compose ps

Docker starts for the first time without starting：

```
ubuntu@ubuntu:/opt/feel_iqueue$ sudo docker-compose ps
Name   Command   State   Ports
------------------------------
```

sudo docker-compose up -d

5.3 Restart IQueue Server

cd /opt/feel_iqueue/

View docker status：
sudo docker-compose ps

If Docker has been started, restart：

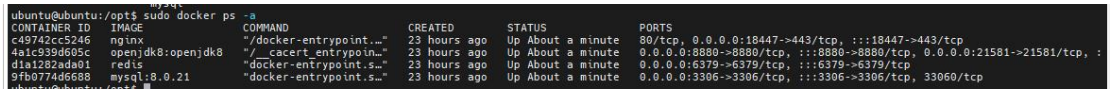
```
ubuntu@ubuntu:/opt/feel_iqueue$ sudo docker-compose ps
   Name                 Command              State                                        Ports
-------------------------------------------------------------------------------------------------------------------------------------------
feel_iqueue   /__cacert_entrypoint.sh ja ...   Up      0.0.0.0:21581->21581/tcp,:::21581->21581/tcp, 0.0.0.0:8880->8880/tcp,:::8880->8880/tcp
queue-web     /docker-entrypoint.sh ngin ...   Up      0.0.0.0:18447->443/tcp,:::18447->443/tcp, 80/tcp
```

sudo docker-compose restart

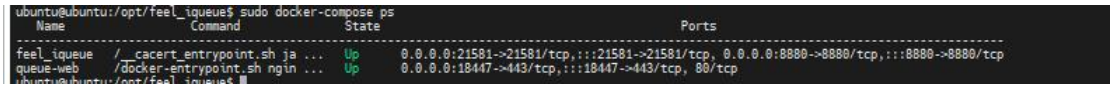
5.4 Restart Linux System

cd /opt/feel_iqueue/

View docker status：
sudo docker ps -a

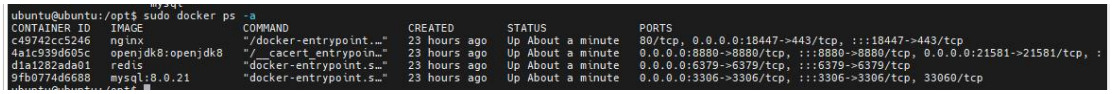


View docker status：
sudo docker-compose ps



Reboot Linux System:
sudo reboot

View docker status, it will auto restart：
sudo docker ps -a



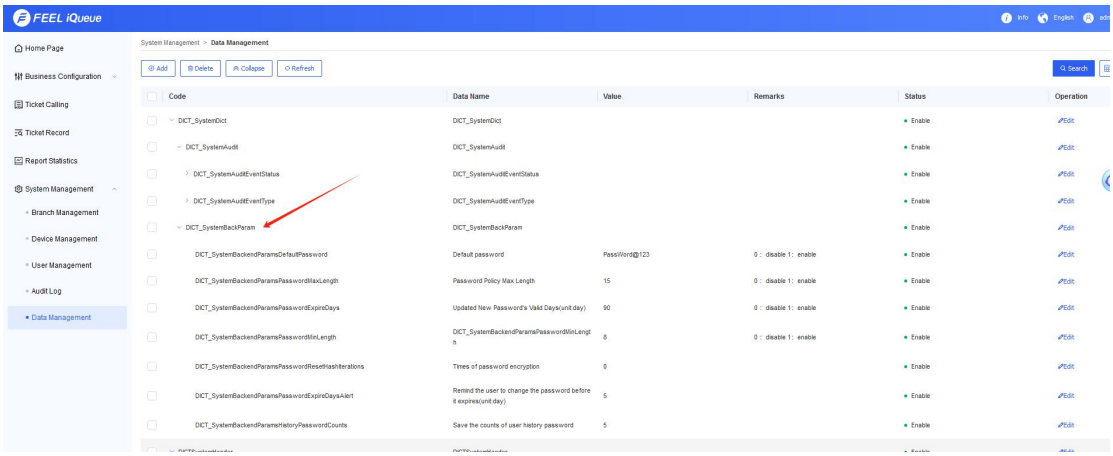
6. IQueue config

6.1 Configuration on the system software

6.1.1 Password Config

Configuration Name	Description
DICT_SystemBackendParamsDefaultPassword	Default password
DICT_SystemBackendParamsPasswordMaxLength	Password Policy Max Length
DICT_SystemBackendParamsPasswordExpireDays	Updated New Password's Valid Days(unit:day)

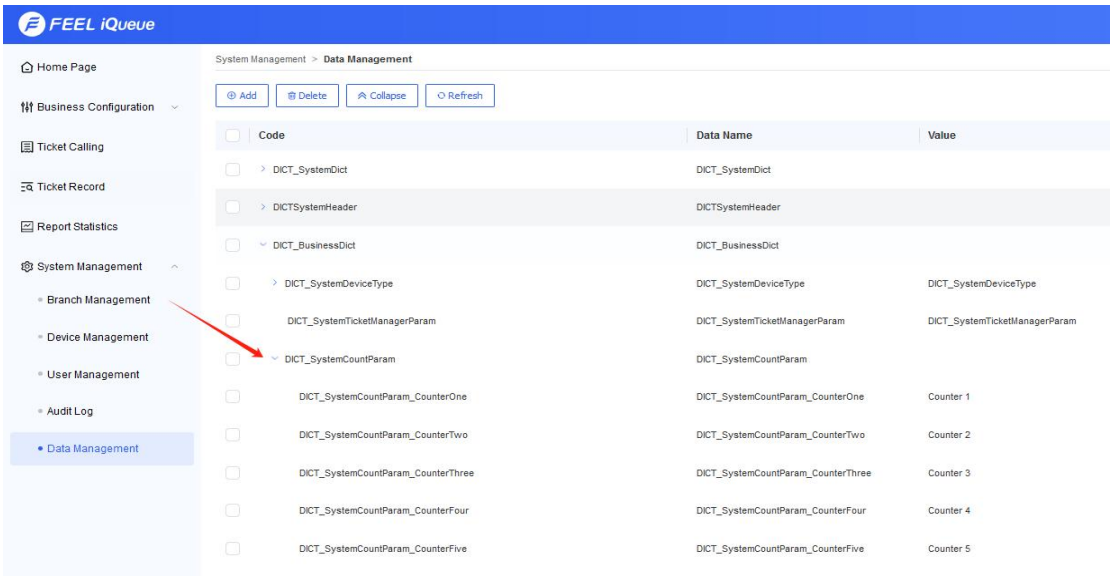
DICT_SystemBackendParamsPasswordMinLength	Password Min Length
DICT_SystemBackendParamsPasswordExpireDaysAlert	Remind the user to change the password before it expires(unit:day)
DICT_SystemBackendParamsHistoryPasswordCounts	Save the counts of user history password
DICT_SystemBackendParamsPasswordResetHashIterations	Times of password encryption，No need to change.



6.1.2 Counter Config

Format requirement：DICT_SystemCountParam_Counter+One/Two/FiftyOne/SixtyOne...

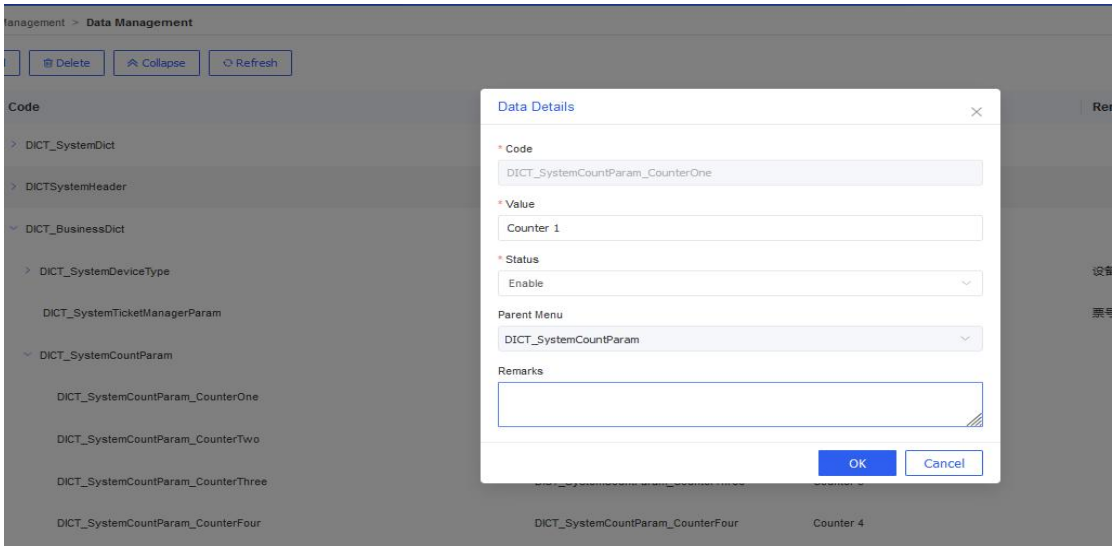
Configuration Name	Description
DICT_SystemCountParam_CounterOne	Counter 1
DICT_SystemCountParam_CounterTwo	Counter 2



New window：

The format must be strictly as follows：

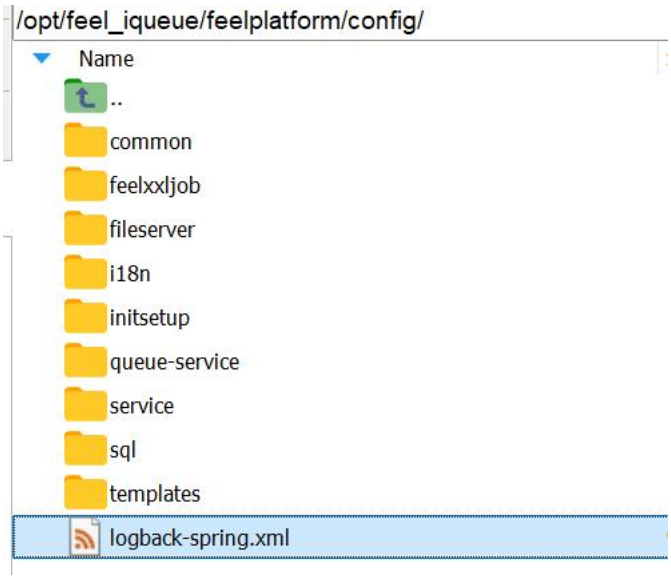
Code	DICT_SystemCountParam_Counter+One/Two/FiftyOne/SixtyOne...
Value	Counter+数字，Counter 1/Counter 2/Counter 11/Counter 110...
Status	Enable
Parent Menu	DICT_SystemCountParam



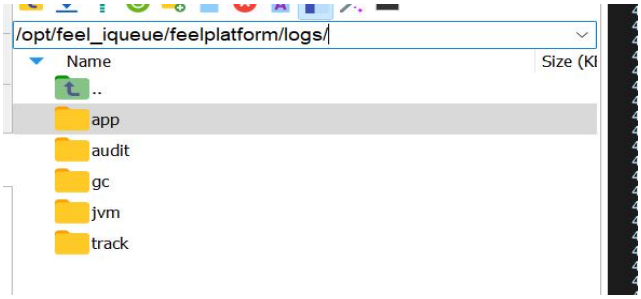
6.2 Configuration file on linux

6.2.1 Logs

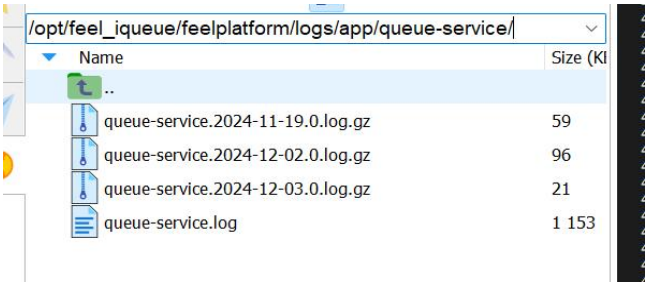
Log configuration directory： /opt/feel_iqueue/feelplatform/config/logback-spring.xml



Log directory：
/opt/feel_iqueue/feelplatform/logs/

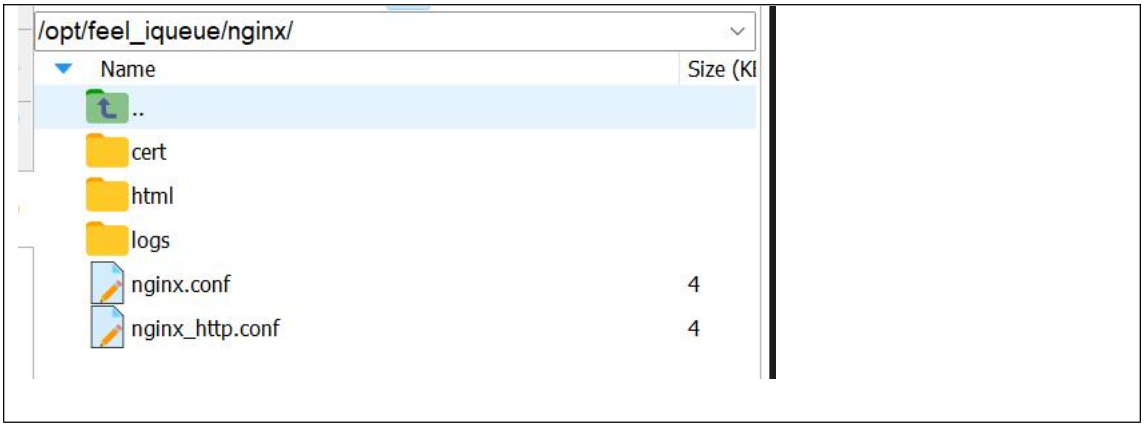


Locating problem directory：
/opt/feel_iqueue/feelplatform/logs/app/queue-service/
/opt/feel_iqueue/feelplatform/logs/track/



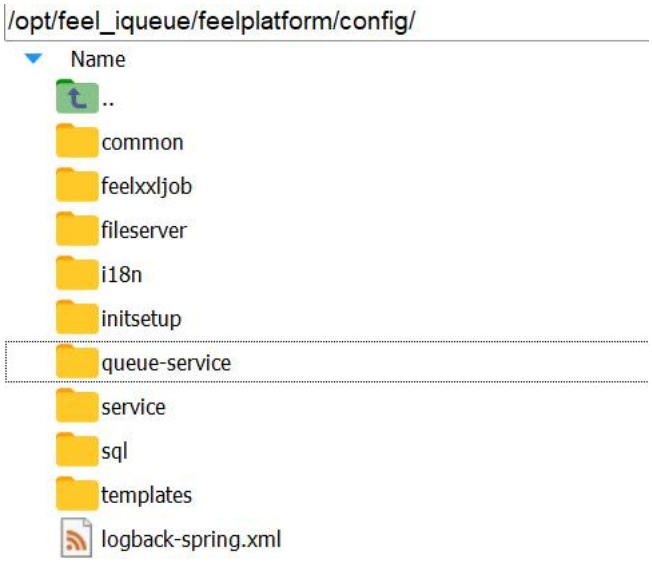
6.2.2 nginx Config

Nginx directory:：
/opt/feel_iqueue/nginx/nginx.conf

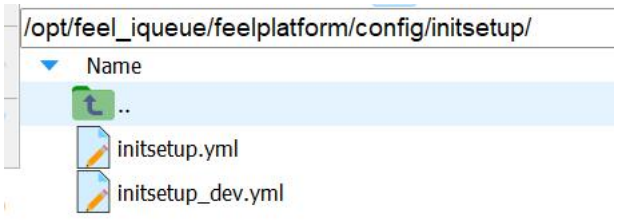


6.2.3 iQueue Config

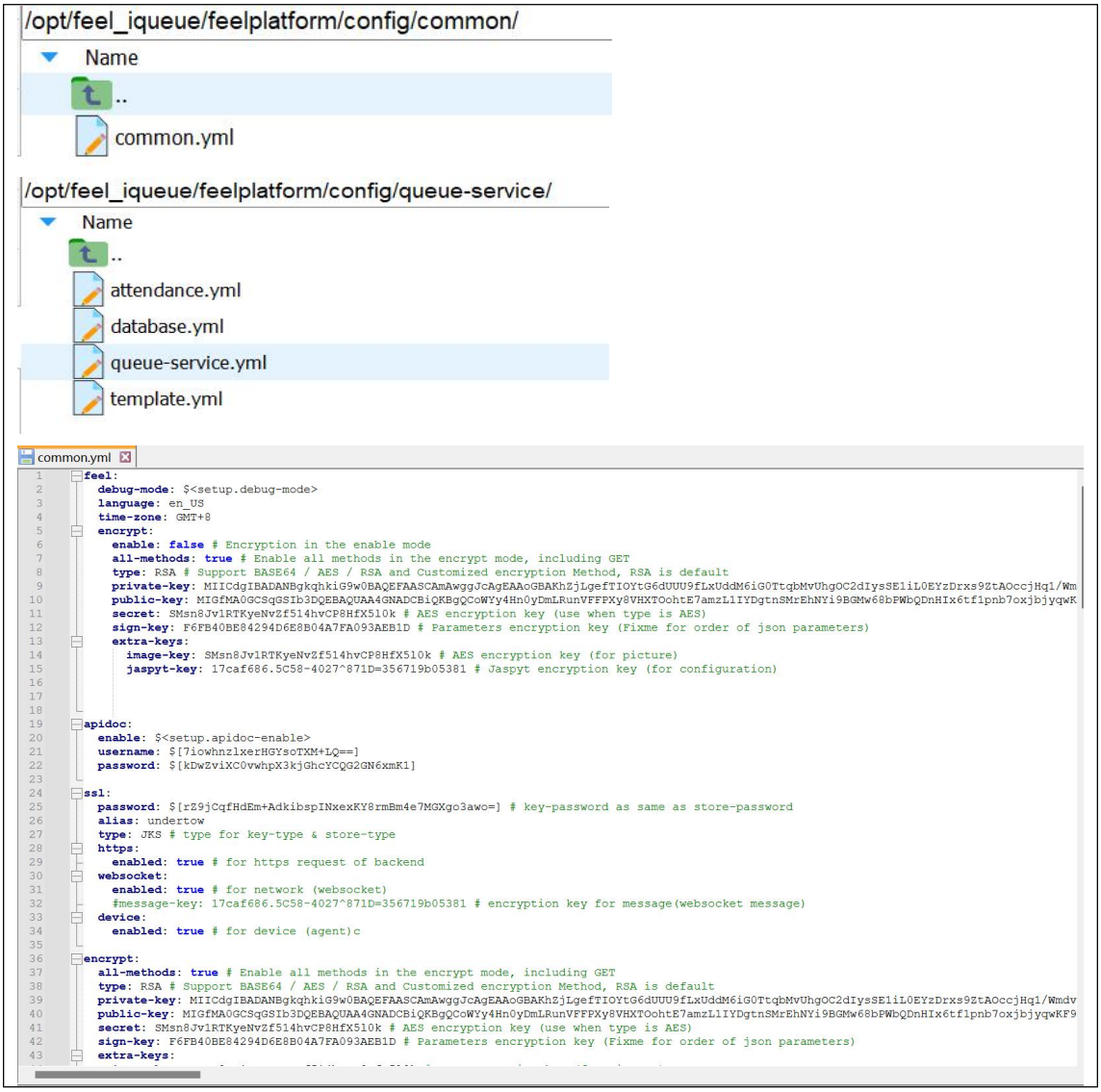
Config directory:
/opt/feel_iqueue/feelplatform/config/



配置 mysql 和 redis 的 IP 地址：
/opt/feel_iqueue/feelplatform/config/initsetup/

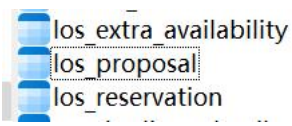


项目相关配置：
/opt/feel_iqueue/feelplatform/config/common/
/opt/feel_iqueue/feelplatform/config/queue-service/



7. Database Support

7.1 LOS Table



Start mysql:

root/Root002152

feelaccess/Feelaccess002152

los_extra_availability：

los_extra_availability @feel_queue_a...

开始事务 | 文本 | 筛选 | 排序 | 导入 | 导出 | 数据生成 | 创建图表

ID	ORGANIZATION_ID	RESERVATION_TIME_	RESERVATION_TIME_	EXTRA_AVAILABILITY	CREATE_BY	UPDATE_BY	CREATE_TIME	UPDATE_TIME
1	100	2024-12-02 08:00:00	2024-12-02 09:00:00	5	(Null)	(Null)	2024-12-02 13:52:16	(Null)
2	100	2024-12-02 09:00:00	2024-12-02 10:00:00	5	(Null)	(Null)	2024-12-02 13:52:16	(Null)
3	100	2024-12-02 10:00:00	2024-12-02 11:00:00	5	(Null)	(Null)	2024-12-02 13:52:16	(Null)
4	100	2024-12-02 11:00:00	2024-12-02 12:00:00	5	(Null)	(Null)	2024-12-02 13:52:16	(Null)

los_proposal：

los_proposal @feel_queue_anxin (lo...

开始事务 | 文本 | 筛选 | 排序 | 导入 | 导出 | 数据生成 | 创建图表

ID	HK_ID	PROPOSAL_NUMBER	PROPOSAL_CHECK_DIGIT	CUSTOMER_NAME	CUSTOMER_DOB	CUSTOMER_GENDER	CUSTOMER_NATIONALITY	CUS
3608082433	9ddababede:	81131270	01	6E34F3BD30DF06E1F	30AFE452A927798F423BF276E1C106D0FEC	4C2852A895091D6BEFBBF	31D4B	
2214794242	4caa7b8dc2f	21817657	20	6E34F3BD30DF06E1F	30AFE452A927798F423BF276E1C106D0FEC	87F4441AF90D63CEC08979	7D4B	
6222211074	c53a5c57eb9	20095748	20	6E34F3BD30DF06E1F	30AFE452A927798F423BF276E1C106D0FEC	87F4441AF90D63CEC08979	7D4B	

los_reservation：

los_reservation @feel_queue_anxin (...

开始事务 | 文本 | 筛选 | 排序 | 导入 | 导出 | 数据生成 | 创建图表

ID	BUSINESS_ID	ORGANIZATION_ID	TIMESLOT_START	TIMESLOT_END	APPOINTMENT_DATETIME	STATUS	PROPOSAL_ID	CREATE_BY	UPDATE_BY
777409	301	100	2024-11-14 14:30:0	2024-08-31 15:30:(	2024-11-14 09:32:00	1	51503608082433	device	device
534082	301	100002	2024-11-14 14:30:0	2024-08-31 15:30:(	2024-11-14 09:32:00	1	51503608082433	device	device

7.2 Ticket record table

queue_ticket_record：

queue_ticket_record @feel_queue_a...

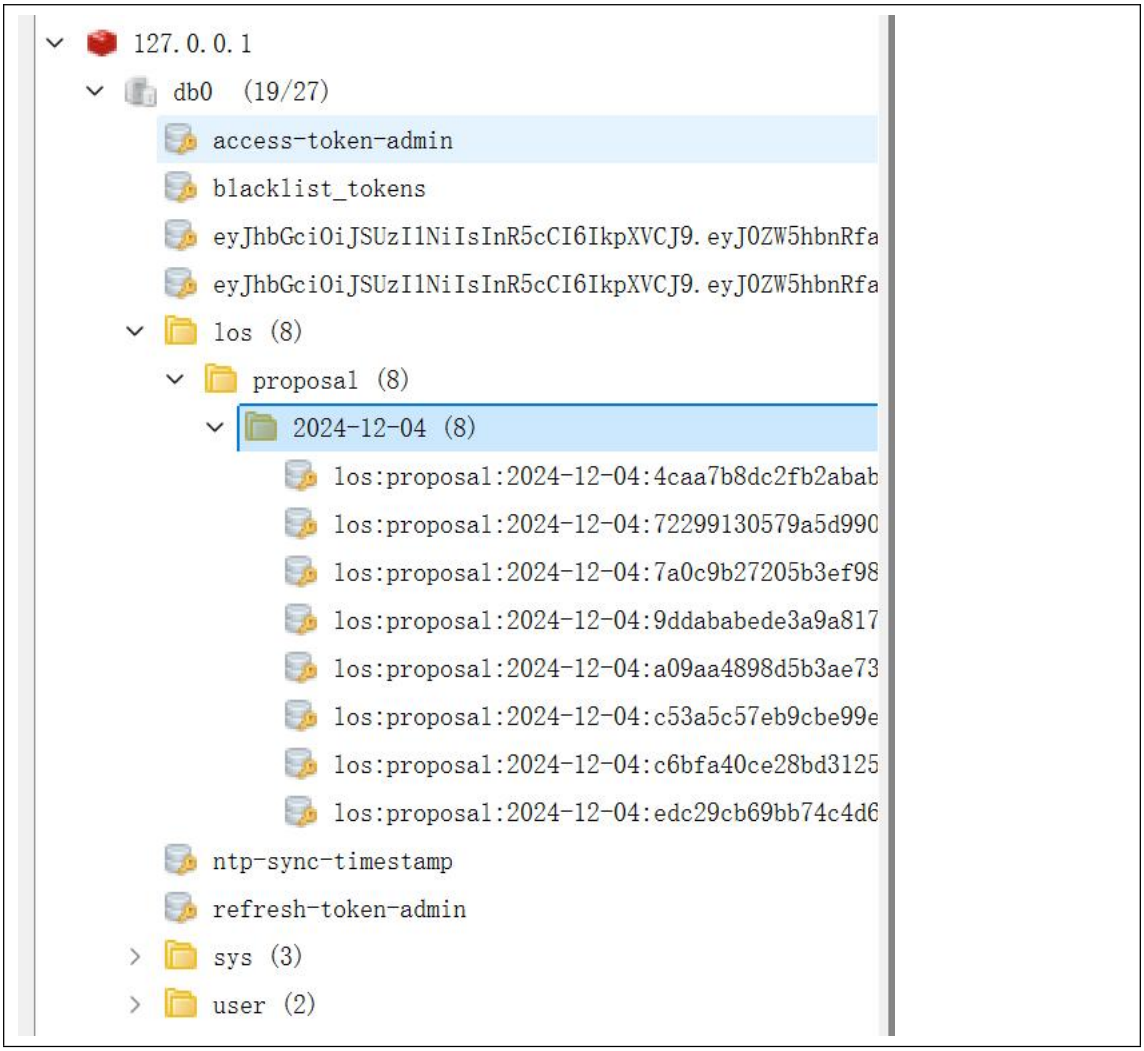
开始事务 | 文本 | 筛选 | 排序 | 导入 | 导出 | 数据生成 | 创建图表

ID	TICKET_NUMBI	BUSINESS_IE	ORGANIZATION	ID_RESERVATIC	APPOINTMENT_[	TICKET_STATU	TICKET_PRINT_ST	TICKET_LOS_CO	LOS_RESERVATIC	ASSIGNED_STA
8904253104130	81131270	301	100	1863461504422	2024-11-14 09:3.0	0	N	2	0	

8. Redis Support

Start redis:

6379: P@ssw0rd


127.0.0.1
db0 (19/27)
access-token-admin
blacklist_tokens
eyJhbGciOiJSUzI1NiIsInR5cCI6IkpXVCJ9.eyJ0ZW5hbnRfa
eyJhbGciOiJSUzI1NiIsInR5cCI6IkpXVCJ9.eyJ0ZW5hbnRfa
los (8)
proposal (8)
2024-12-04 (8)
los:proposal:2024-12-04:4caa7b8dc2fb2abab
los:proposal:2024-12-04:72299130579a5d990
los:proposal:2024-12-04:7a0c9b27205b3ef98
los:proposal:2024-12-04:9ddababede3a9a817
los:proposal:2024-12-04:a09aa4898d5b3ae73
los:proposal:2024-12-04:c53a5c57eb9cbe99e
los:proposal:2024-12-04:c6bfa40ce28bd3125
los:proposal:2024-12-04:edc29cb69bb74c4d6
ntp-sync-timestamp
refresh-token-admin
sys (3)
user (2)